Module 0: Introduction to AI

Q1. Which type of AI is specifically engineered and limited to performing a single, predefined task?

- A) General AI
- B) Super AI
- C) Narrow (Weak) AI **
- D) Strong AI

Q2. What form of AI remains entirely hypothetical and raises significant existential and ethical questions?

- A) Narrow AI
- B) Super AI **
- C) Applied AI
- D) Weak AI

Q3. Complete the correct nesting hierarchy from the outermost discipline to the inner sub-specialization:

- A) Deep Learning $\subset$ Machine Learning $\subset$ Artificial Intelligence
- B) Artificial Intelligence $\subset$ Deep Learning $\subset$ Machine Learning **
- C) Artificial Intelligence $\subset$ Machine Learning $\subset$ Deep Learning
- D) Machine Learning $\subset$ Artificial Intelligence $\subset$ Deep Learning

Q4. Which lifecycle framework outlines phases like business understanding, data preparation, modeling, and deployment?

- A) Agile Scrum
- B) CRISP-DM **
- C) DevOps Pipeline
- D) Six Sigma

Q5. Machine Learning allows computers to learn from data and improve performance without being:

- A) Fed with matrices
- B) Explicitly programmed **
- C) Monitored by engineers
- D) Connected to a network

Module 1: Supervised Learning

Q6. What characterizes Supervised Learning?

- A) The data does not include labels.
- B) The model learns from dynamic environmental penalties without data.

- C) The algorithm is trained on a labeled dataset containing input-output pairs. **
- D) The model only groups elements by raw similarity metrics.

Q7. Which task aims to predict a continuous numerical value?

- A) Classification
- B) Clustering
- C) Regression **
- D) Association Rules

Q8. In the simple linear regression formula $y = \beta_0 + \beta_1x + \epsilon$, what does β_1 represent?

- A) The dependent target variable
- B) The intercept
- C) The random error term
- D) The slope **

Q9. What is the standard loss function minimized during classic linear regression training?

- A) Cross-Entropy Loss
- B) Hinge Loss
- C) Mean Squared Error (MSE) **
- D) Absolute Mean Percentage Error

Q10. What is the fundamental objective of a Support Vector Machine (SVM) in binary classification?

- A) To minimize cluster centroid distances.
- B) To find the optimal hyperplane that maximizes the margin between classes. **
- C) To sequence words in a backward directional loop.
- D) To count transition probabilities.

Q11. Why does a Soft-Margin SVM introduce slack variables (ξ_i)?

- A) To project data into infinite-dimensional matrices.
- B) To allow the network to handle text sequences.
- C) To handle data that is not perfectly linearly separable by penalizing errors via parameter C. **
- D) To calculate the variance of data attributes.

Q12. What technique do SVMs utilize when data is non-linearly separable in its original space?

- A) Experience Replay
- B) The Kernel Trick **
- C) Max Pooling
- D) Backpropagation Through Time

Q13. Which of the following is NOT a popular kernel function used in SVM configurations?

- A) Radial Basis Function (RBF)
- B) Polynomial Kernel
- C) Sigmoid Kernel
- D) Softmax Kernel **

Q14. What are the three standard sequential layers found in an Artificial Neural Network (ANN)?

- A) Input Layer, Hidden Layer(s), Output Layer **
- B) Convolutional Layer, Pooling Layer, Fully Connected Layer
- C) Replay Layer, Target Layer, Action Layer
- D) Support Layer, Vector Layer, Hyperplane Layer

Q15. Given input features x_i, weights w_i, and bias b, what formula does a single neuron calculate before activation?

- A) $z = \max(0, x \cdot w)$
- B) $z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$ **
- C) $z = \frac{1}{1 + e^{-x}}$
- D) $z = \sum (y_i - \hat{y}_i)^2$

Q16. Which activation function scales values strictly to a probability distribution between 0 and 1?

- A) ReLU
- B) Tanh
- C) Sigmoid **
- D) Linear

Q17. What mathematical output is yielded by a standard ReLU activation function if the incoming value z is negative?

- A) -1
- B) 1
- C) 0 **
- D) An exponentially shrinking decimal

Q18. What is the purpose of backpropagation in an ANN?

- A) To initialize weights completely at random.
- B) To calculate the gradients of the loss function to update the model weights. **
- C) To map text characters into transactional basket itemsets.
- D) To strip raw data of its continuous attributes.

Q19. Which loss function is most appropriate for a multi-class neural network classification task?

- A) Mean Squared Error
- B) Cross-Entropy Loss **
- C) Mean Absolute Deviation
- D) Root Mean Squared Error

Q20. Algorithms like Gradient Descent or Adam are primarily used in ANNs for:

- A) Feature extraction
- B) Optimization during training **
- C) Normalizing image dimensions
- D) Designing Dendrogram structures

Module 2: Unsupervised Learning

Q21. What happens in Unsupervised Learning?

- A) Data contains clear, labeled ground-truth answers.
- B) The model infers internal groupings or structural patterns directly from unlabeled data.**
- C) An agent interacts with a physical environment via reward tables.
- D) A supervisor corrects every wrong output live during training.

Q22. K-Means is categorized as what type of clustering approach?

- A) Soft Clustering
- B) Fuzzy Clustering
- C) Hard Clustering **
- D) Hierarchical Dendrogram Clustering

Q23. What objective is minimized during the iteration loops of the K-Means algorithm?

- A) The global margin of hyperplanes.
- B) The variance/squared Euclidean distances within each cluster to its centroid. **
- C) The number of components selected.
- D) The confidence score of rules.

Q24. What visual tree structure is built when executing Hierarchical Clustering?

- A) Decision Tree
- B) Replay Buffer Graph
- C) Dendrogram**
- D) Neural Network Vector Plot

Q25. Which linkage criterion calculates the maximum distance between points in two distinct clusters?

- A) Single Linkage
- B) Average Linkage
- C) Complete Linkage **
- D) Centroid Linkage

📝 Part 2: Questions 26 – 50

Q26. Principal Component Analysis (PCA) is primarily utilized for:

- A) Predicting real estate market numbers.
- B) Dimensionality reduction while preserving as much variance as possible. **
- C) Categorizing items into discrete binary classes.
- D) Simulating environment rewards for robotic components.

Q27. Why must data be standardized before executing PCA?

- A) To force text datasets to drop long sentence structures.
- B) To ensure features with larger raw units do not disproportionately dominate the variance calculation. **
- C) To replace zero values with random positive integers.
- D) To compute confidence boundaries for transaction subsets.

Q28. What do the eigenvectors derived from a covariance matrix represent in PCA?

- A) The magnitude of error metrics.
- B) The directions of maximum variance in the data space. **
- C) The frequency of itemsets appearing together.
- D) The specific nodes within hidden layers.

Q29. What matrix represents how data features co-vary with each other during a PCA workflow?

- A) Identity Matrix
- B) Replay Matrix
- C) Covariance Matrix **
- D) Confusion Matrix

Q30. Which algorithm operates on the principle that any subset of a frequent itemset must also be frequent?

- A) Q-Learning
- B) K-Means
- C) Apriori Algorithm **
- D) Support Vector Machine

Q31. In association rules, what metric defines the overall proportion of total transactions containing an itemset A?

- A) Confidence
- B) Support **
- C) Lift
- D) Slack

Q32. For an association rule $A \implies B$, what does a Lift score exactly equal to 1 indicate?

- A) A and B are completely independent of each other. **
- B) B is perfectly caused by A.
- C) No transaction contains both items.
- D) The confidence of the rule is exactly 0.

Q33. If Support($A \cup B$) is 0.3 and Support(A) is 0.6, what is the Confidence of the rule $A \implies B$?

- A) 0.18 **
- B) 2.0
- C) 0.5 (calculated as $0.3 / 0.6$)
- D) 0.9

Module 3: Deep Learning Architectures

Q34. What deep learning architecture is uniquely specialized for processing grid-like data inputs like images?

- A) Recurrent Neural Network
- B) Long Short-Term Memory Network
- C) Convolutional Neural Network **
- D) Tabular Perceptron

Q35. What operation occurs inside a CNN's convolutional layer?

- A) Randomly dropping weights to clear context memory.
- B) Sliding a filter/kernel over the input to compute dot products and extract feature maps.**
- C) Flattening a 2D matrix into a 1D sequential vector array.
- D) Selecting actions to optimize environment rewards.

Q36. What is the main objective of a pooling layer in a CNN?

- A) To add non-linearity using a Sigmoid curve.
- B) To downsample spatial dimensions, reducing dimensionality and computational load.**

- C) To calculate error loss relative to standard labels.
- D) To update backpropagation weights using temporal chains.

Q37. What type of pooling operation extracts the absolute highest numerical value within a designated matrix window?

- A) Average Pooling
- B) Global Mean Pooling
- C) Max Pooling **
- D) Stochastic Linkage Pooling

Q38. What structural components at the end of a CNN architecture combine all extracted features for final classification?

- A) Convolutional Layers
- B) Fully Connected Layers **
- C) Recurrent Forget Gates
- D) Linear Covariance Vectors

Q38. Which neural architecture is specifically optimized for handling sequential data streams and text analysis?

- A) Hard Margin SVM
- B) Standard Feedforward Network **
- C) Recurrent Neural Network (RNN)
- D) Principal Component Analyzer

Q39. What structural element acts as an internal memory for basic RNN architectures across temporal sequences?

- A) Slack Variable
- B) Hidden State (h_t)
- C) Kernel Filter Matrix **
- D) Reward Tensor

Q40. What critical training limitation occurs when computing gradients across long input text sequences in basic RNNs?

- A) The Vanishing/Exploding Gradient Problem
- B) Over-clustering of components **
- C) Loss of the Covariance Matrix
- D) Inseparable hyperplanes

Q41. Which advanced variants of RNNs were designed to actively solve the vanishing gradient problem?

- A) Linear SVM and Soft SVM **

- B) K-Means and Hierarchical Clustering
- C) LSTM and GRU
- D) PCA and Apriori

Q42. Which gate within an LSTM cell determines exactly what slice of historical context information to discard?

- A) Input Gate
- B) Forget Gate
- C) Output Gate **
- D) Replay Gate

Module 4: Reinforcement Learning

Q43. In Reinforcement Learning, who or what is the active learner/decision-maker?

- A) The Environment
- B) The Agent **
- C) The Policy
- D) The Replay Buffer

Q44. Which term defines the strategy or mapping system that an agent employs to choose actions based on current states?

- A) Value Function
- B) Reward Signal **
- C) Policy (π)
- D) Eigenvector

Q45. Q-Learning is explicitly defined as what type of machine learning algorithm?

- A) Supervised Continuous Regression
- B) Model-free, off-policy Reinforcement Learning
- C) Unsupervised Dimensionality Reduction **
- D) Deep Convolutional Image Segmenter

Q46. What does the Q-function, $Q(s,a)$, represent in an RL system?

- A) The current coordinate location of the network architecture.
- B) The expected cumulative future return of taking action a in state s. **
- C) The absolute error value of an optimization matrix.
- D) The count of transaction itemsets across database rows.

Q47. In the Q-Learning update rule, what parameter is represented by γ ($0 \le \gamma \le 1$)?

- A) The learning rate
- B) The slack variable multiplier **

- C) The discount factor for future rewards
- D) The kernel dimension filter

Q48. Why is classical tabular Q-Learning limited or infeasible for complex domains like modern video games?

- A) It requires labeled images of every move.
- B) It cannot scale to large or continuous high-dimensional state-action spaces.
- C) It does not use any optimization calculus formulas.**
- D) It is a supervised approach dependent on human labels.

Q49. How does a Deep Q-Network (DQN) scale up the capacity of classical Q-Learning?

- A) It eliminates rewards completely.
- B) It uses a deep neural network to approximate the Q-function values. **
- C) It limits actions exclusively to a single binary boundary.
- D) It converts the data layout back into market item rule sets.

Q50. What innovation is used in DQN to break correlations between sequential training data?

- A) Fixed Target Networks
- B) Experience Replay Buffer **
- C) Polynomial Kernel Projection
- D) Hierarchical Completeness Linkage